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Temperature response functions for residential energy demand – A review of models

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ABSTRACT

Climate is a major influence on residential energy demand and energy demand patterns. A better understanding of the link between climate and energy demand responses is crucial to improve energy forecasting outcomes and to evaluate the potential of climate change adaptation strategies. This paper reviews models of residential energy demand to temperature. We classify the models used in papers published in peer-reviewed journals and observe that the credibility of energy demand projections hinges in part on the assumption that the shape and type of the temperature response function is stationary. The evidence of acclimatization to warmer conditions may oppose this assumption. While the studies reviewed here only tested the impact of changes in the temperature thresholds and the shape of temperature response functions, a comprehensive understanding of changes in energy demand patterns and climate adaptation potentials requires knowledge of building stock turnover rates and associated changes in technology and behavioral responses of building occupants. These are not yet part of the standard repertoire of temperature-based demand models.

Contents

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1. Introduction

The impact of changes in energy use on greenhouse gas emissions has long occupied the climate change research community (Hillman and Ramaswami, 2010; Lenzen, 1998; Schipper et al., 1997). Similarly, climate impacts on energy use and electricity load are well researched, showing that temperatures outside desirable ranges call for changes in heating and cooling demand. What that relationship between temperature on the one hand, and heating and cooling demand on the other hand looks like, and how it unfolds as climate changes, however, is significantly less understood. Yet, temperature-energy demand relations are used to project, seasonally and over the long term, the capacity requirements of the energy sector, fuel use in electricity generation, space heating and cooling needs by final consumers, and associated emissions (e.g. De Cian et al., 2007; Kirshen et al., 2008; Mansur et al., 2005; Moral-Carcedo and Vicéns-Otero, 2005).

Energy demand is typically more responsive in the residential sector to weather changes than in the commercial sectors, in part because the latter are dominated by process needs and the smaller ratio of building envelope surface area to interior space (Brown et al., 2015). Besides, space conditioning in commercial settings is less adjusted over the course of any given day. The relative importance of temperature for energy demand in the residential sector therefore makes an understanding of temperature-energy demand relationships of particular importance for energy sector management in urban areas. The higher the impact of temperature on demand, for example, the more critical it will be to plan for capacity changes and deploy demand side management measures as climate change accentuates demand profiles. Additionally, the more geographically concentrated are the demand change, the more affected can be the transmission and distribution networks on which energy demand depends.

While researchers have studied the influence of other potential factors such as population, household size, income, price of fuel and building characteristic on residential energy demand, this paper focuses on various temperature response functions (TRF) that have been used in previous studies to quantify and model how residential energy demand varies with temperature. A central concept in these studies is the use of degree day calculations (Madlener and Alt, 1996; Parti and Parti, 1980), where a degree-day is defined as the difference between a day's average temperature and some temperature threshold, often referred to as the balance point temperature (T_b):

$$\mathrm{HDD}_T = \sum_{days} (T - T_{b,h}) \quad \text{when } T \geqslant T_{b,h} \tag{1}$$

$$\mathrm{CDD}_T = \sum_{days} (T_{b,c} - T) \quad \text{when } T \leqslant T_{b,c} \tag{2}$$

In the above equations, T is the daily average temperature of a region and $T_{b,h}$ and $T_{b,c}$ are the balance point temperatures for heating and cooling, respectively.

Where there is insufficient daily data or where research interests are on broader aggregate trends, average monthly temperatures are used (Asadoorian et al., 2008; De Cian et al., 2012; Fung et al., 2006; Mansur et al., 2008; Mendelsohn, 2003; Moral-Carcedo and Vicéns-Otero, 2005). Whether for short or long-term assessments, there are methodological and data challenges surrounding the choice of balance point temperature(s), assumptions about the speed at which households switch between heating and cooling, and assumptions about changes over time in thresholds and demand behaviors that may come from changes in technology and preferences.

In this paper, we primarily review the best-fitting short-run response functions relating HDD and CDD to residential energy demand. In the subsequent section, we review the application of TRFs in energy demand modeling and forecasting. We begin with assumptions of a single balance point temperature and linear, symmetric responses around it, and eventually attend to studies dealing with multiple thresholds and with non-linear and asymmetric responses. We close our review with a discussion of the potential for the respective specifications to help evaluate the capacity of adaptation to climate change for residential energy demand. As the evaluation of adaptation strategies depends directly on how the climate sensitivity of residential cooling and heating energy use is modeled, we recognized gaps in the literature that, if closed, could help strengthen the scientific basis for such strategies.

2. Temperature response functions

This study expands a recent review by Auffhammer and Mansur (2014), which focused on cross-sectional climatic evidence and panel (or time series) evidence of weather shocks, and Brown et al. (2015) which classified studies on the basis of models used in analysis. Here, we group the literature by types of TRF used to model relationships between climate and energy demand in the residential sector, and compare the abilities of these functions to predict energy use and to assess impacts of climate adaptation measures.

2.1. Linear symmetric models

Linear symmetric models are based on the assumption that there is a single balance point temperature for heating and cooling (Fig. 1). Additionally, some studies assume symmetric sensitivity to a marginal change in temperatures around the

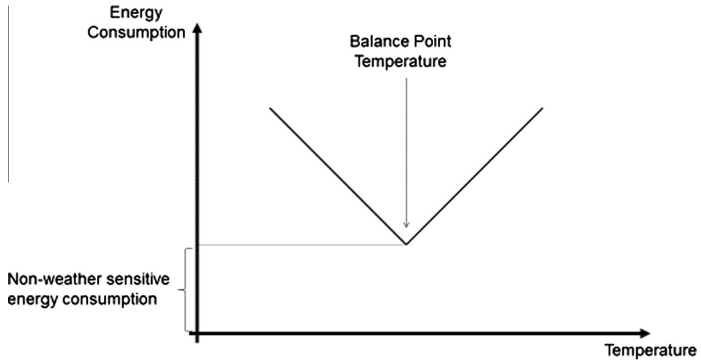

Fig. 1. Linear symmetric model for a relationship between temperature and energy consumption.

balance point. The outcome is a V-shaped relationship between temperature and energy use (Jager, 1983). This model has generally been used in sophisticated building model simulations (Day, 2006) and in utility demand forecasting models (Engle et al., 1986). Its appeal lies in the simplicity of its specification and empirical estimation, and the ease with which findings can be compared across geographies and time, for example.

Early energy studies, focused on the US, simply consider 65 °F (18.3 °C) as the balance point for the space conditioning temperature relationship (Considine, 2000; de Dear and Brager, 2001; Donovan and Fischer, 1976; Lam, 1998; Nall and Arens, 1979; Pardo et al., 2002; Quayle and Diaz, 1980; Valor et al., 2001) . Others, in global analyses, use 18 °C for both heating and cooling (Isaac and van Vuuren, 2009; Labriet et al., 2013). However, the actual balance point temperature depends on place-specific characteristics of the building stock, non-temperature weather factors (e.g., humidity, precipitation, and wind), and cultural preferences and practices (dress codes, siestas, and such). Therefore, recent energy demand analysis has attempted to find the appropriate balance point temperature for specific regions and sectors. For example, Sailor and Munoz (1997) found a balance point of 18.3 °C for both heating and cooling for all U.S. states, except for the State of Florida (USA), where it was estimated to be 21 °C. Amato et al. (2005) estimated for Massachusetts, USA, balance point temperatures for electricity consumption in the residential and commercial sectors, respectively, of 60 °F (15.6 °C) and 55 °F (12.8 °C). Sarak and Satman (2003) in a study for Turkey identified 15 °C for natural gas use for heating while Giannakopoulos and Psiloglou (2006) used a balance point temperature of 22 °C for electricity use in Athens, Greece. More recently, Vu et al. (2015) used 19.5 °C as the balance point temperature to estimate the monthly electricity demand based on the temperature data for the years 1999–2010 for the state of New South Wales, Australia. Heim et al. (2003) proposed a residential energy demand temperature index (REDTI), based on total population-weighted heating and cooling degree days (base 18.3 °C) in the contiguous United States, to describe the sensitivity of present-day residential heating and cooling energy-use patterns to variations in climate.

While the majority of studies presumed a balance point temperature – typically at or close to 65 °F (18.3 °C) – several studies explored the potential ability to explain more of the variation in their data by choosing different balance point temperatures. One such approach is to iteratively run a set of statistical models over a range of balance point temperatures (at 1 or 2 °F intervals), to identify the one that explains the largest share of changes in energy use for each energy carrier in each end use sector (Kissock et al., 2003; Ruth and Lin, 2006; Timmer and Lamb, 2007).

More recently, Cho et al. (2011) suggest a modified model to estimate HDDs by defining an unloaded temperature, which is the balance point temperature corrected for the effects of variable solar and internal heat gains, as shown in Eq. (3). As a consequence of that correction, the unloaded temperature is smaller than the balance point temperature.

$$\mathrm{HDD}_{T_i - T_z} = m_k \sum_{k=1}^{n} (T_i - T_o) + m_k \sum_{k=1}^{n} (T_i - T_z) \tag{3}$$

In the above equations, T_o is the daily average temperature of a region while T_i is the balance point temperature and T_z the unloaded temperature. The coefficient m_k has the value of 1 (or 0) if the daily average temperature (T_o) is lower (the same as or higher) than the unloaded temperature (T_z). Cho et al. (2011) applied equation (3) to 15 cities across Korea, and concluded that as the revised approach accounts for the internal heat gain and that it is more practical compared to the HDD method based only on the standard use of a balance point temperature of Eq. (1) (Fig. 2).

2.2. Linear asymmetric models

One common assumption inherent in all the linear symmetric models is that there is a shift from heating devices to cooling equipment for infinitesimal deviations from the balance point temperature, which is not realistic. Consequently, several studies intended at refining the linear V-shaped models by specifying a comfort zone (Fig. 3), which is the temperature range at which no heating or cooling is required to keep the indoor temperature at the required level (Eskeland and Mideksa, 2010;

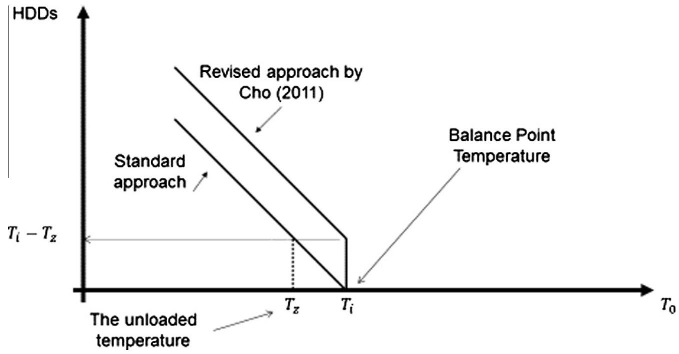

Fig. 2. Comparison of HDD calculated from standard approach and the revised approach by Cho et al. (2011).

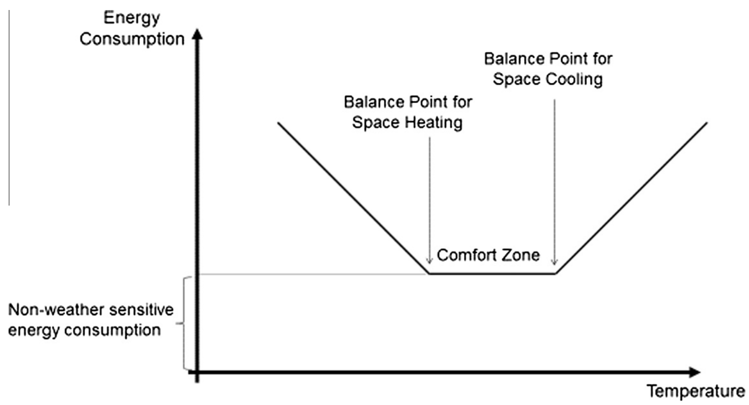

Fig. 3. Linear model with a comfort zone for the relationship between temperature and energy consumption – Adjusted from Brown et al. (2015).

Hekkenberg et al., 2009; Moral-Carcedo and Vicéns-Otero, 2005; Shorr et al., 2009). Presence of a comfort zone may be the product of preferences, but also of technology, such as building shell characteristics, that do not lead occupants to adjust indoor temperatures as outdoor conditions vary within a more limited range than outdoor conditions.

Hamlet et al. (2010) used 18.3 °C and 23.9 °C as the temperature threshold for calculating HDDs and CDDs to project climate change impacts on energy supply and demand in the Pacific Northwest and Washington State (USA). While recognizing the fact that energy consumption does not instantaneously increase with even marginal deviations from a single balance point, this approach retains some of the elegance and ease of empirical applications that have been a hallmark of other studies using linear temperature-energy demand models.

Dubin (2008) found that a balance point temperature of 65 °F (18.3 °C) used for HDD measurement would tend to overestimate the amount of energy requirement for space heating in the Puget Sound region (Washington State, USA). Therefore, he estimated the balance point temperature based on the thermal coefficients from the Dubin–McFadden engineering thermal model (Dubin and McFadden, 1984) and thermostat settings for individual buildings. This model requires several inputs to estimate space heating demand for each month that would likely occur for each household, which contain: (i) house square footage, (ii) number of storm windows, (iii) number of non-storm windows, (iv) presence of wall insulation, (v) inches of attic insulation, (vi) number of rooms, (vii) number of floors, (viii) weather conditions for system design (ix) 30-year normal heating and cooling degree days for estimated system coefficients of performance and (x) monthly heating and cooling degree days for the period 2001 through 2003. Based on the model outputs, he added a breakpoint at 45 °F (7.2 °C) so that the leading area up to 65 °F is divided into two sections (Fig. 4).

2.3. Nonlinear models

There are several elements that affect the relationship between the demand for space heating and outside air temperature recognized by Henley and Peirson (1997). First, at a low temperature, consumers' ability to reach a preferred dwelling temperature may be limited by the capacity of the heating system. Secondly, consumers may react to very low temperatures by changing their behavior, for example the wearing of warmer clothes. Henley and Peirson concluded that due to the nonlinearity of these physical, economic and other consumer choice effects on the relationship between the demand for space heating energy and outside air temperature, the suitability of a linear approximation should be the subject of empirical

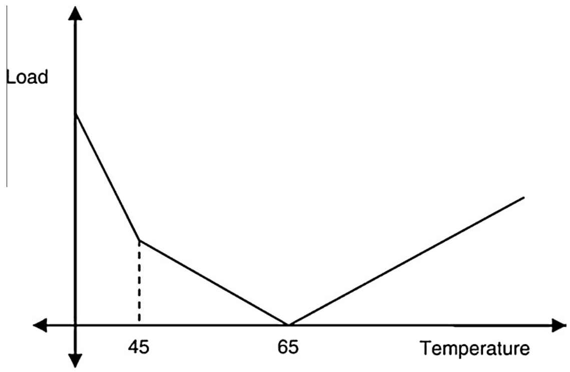

Fig. 4. Temperature–electricity load relationship using multiple balance point temperatures (adapted from Dubin (2008) Fig. 4).

investigation. They found that the polynomial estimation fits better to the data on electrical space heating for a sample of UK households compared to the linear estimation (Henley and Peirson, 1997). However, they found that the results from their polynomial function are biased at extreme temperatures, and may fail to capture threshold effects in behavior.

More recently, Franco and Sanstad (2008) studied time series variation over the course of a year to estimate the impact of temperature on electricity consumption data provided by the California Independent System Operator (CalISO) for 2004. They found a nonlinear influence of average temperature on electricity load, and a linear effect of maximum temperature on peak demand.

The National Energy Modeling System (NEMS) was designed and implemented by the Energy Information Administration (EIA) of the U.S. Department of Energy (DOE). The weather correction formula for the space heating was linear, similar to Eq. (1) for the POLES[1] model. However for space cooling, an exponential weighting was applied to model a non-linear impact of temperature changes on energy use by fuel type (f), building type (b), Census division (r), and year (y), as shown in Eq. (4):

$$\text{Space Cooling Energy Use}_{f,b,r,y} = \text{Space Cooling Energy Use}_{f,b,r,2005} * \left(\frac{\text{Cooling Degree Days}_{r,y}}{\text{Cooling Degree Days}_{r,2005}}\right)^{\alpha} \tag{4}$$

NEMS uses NOAA[2]-supplied data on historical trends to project degree-day values into the future, weighted by population for each census division. As shown in Fig. 5, they used the common 65 °F balance point and the exponent factor estimated to be 1.1 for commercial buildings. Therefore the result is almost non-linear (EIA, 2001).

Brown et al. (2015) followed the approach used in NEMS and found that the best-fitting CDDs have a balance point temperature of 67 °F, for both residential and commercial buildings rather than the conventional 65 °F. When CDDs are based on 65 °F, the best fitting exponent is 1.5 for both residential and commercial buildings. The higher exponent suggests that space cooling is more climate-sensitive than is indicated in NEMS.

Asadoorian et al. (2008) used a log-linear formulation to estimate temperature elasticity, as well as price and income elasticities of electricity demand separately for urban and rural residents for Chinese provinces over a six-year period. They applied two-stage regressions. The first stage estimates the localized impact of climate and other variables on the purchase of electricity-using residential appliances; the second stage then estimates the localized effect of climate and other variables on the intensity of both appliance and non-appliance uses of electricity. They estimated the temperature elasticities of electricity demand to be 0.59 and 0.76 for urban and rural residents, respectively. Besides, the results indicate that price and income elasticities are consistently higher in absolute magnitude with the inclusion of mean temperature variables in the regressions. Moreover, they assessed the significance of including temperature variables for each of the urban and rural residential regressions and found that the cases of excluding mean temperature variables as restricted models was rejected at the 5% level.

Deschênes and Greenstone (2011) documented the relationship between daily temperatures and annual residential energy consumption based on an annual state-level panel data file for total energy consumption from 1968 to 2002 by the residential sector using a U-shaped response function. In their analysis, the total residential energy consumption is comprised of two pieces: "primary" consumption, which is the actual energy consumed by households, and "electrical system energy losses." Also, in addition to estimating the HDD and CDD (using 65 °F as the balance point temperature), they sorted each day's mean temperature experienced by household into nine daily mean temperature bins. Based on their findings on total residential energy consumption measured at the state by year level, the statistical accuracy is an issue in both the degree-days and the "bins" specifications. The results also reveal that the standard approach of modeling energy

[1] Prospective Outlook on Long-term Energy Systems.
[2] The National Oceanic and Atmospheric Administration.

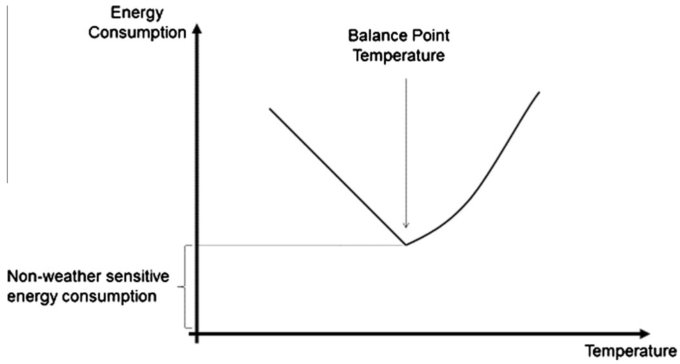

Fig. 5. Non-linear model for relationship between temperature and energy consumption. Adjusted from Brown et al. (2015).

consumption with heating and cooling degree-days hides the nonlinear increase in energy consumption at extremely high temperatures.

Using a global dataset on energy demand for 31 OECD[3] and non-OECD countries, De Cian et al. (2012) established a relationship between residential energy demand and average seasonal temperature using a log-linear formulation. The major contribution of the study was the introduction of a clustering algorithm, which enables the identification of regional heterogeneities. Benefitting from data at global scale, the estimated effects of temperature on energy demand can be used in climate-economy models to assess the overall impacts of climate change on energy demand.

Auffhammer and Aroonruengsawat (2011) examined the impact of climate change on residential electricity consumption at sixteen climate zones[4] using household-level panel data on electricity billing in California. As an original approach to include temperature, they classified each day's mean temperature experienced by household into *k* temperature bins. They accounted for household constant effects, month-of-year fixed effects and geographically specific responses of residential electricity consumption using a simple log-linear specification. They calculated the temperature slope coefficient for each of the fourteen percentile bins, which is approximately equal to the percentage change in household electricity consumption due to experiencing one additional day in each temperature bin for the sixteen climate zones. Their study found that the areas with the sharpest response functions at higher temperature bins are the locations with highest growths in the number of high and extremely high temperature days.

2.4. Semi- or non-parametric models

Instead of assuming linearity in HDD and CDD, Engle et al. (1986) applied a semi-parametric regression procedure to estimate the relationship between electricity load and temperature. A customer's income, energy price and monthly indicators entered the model linearly, while weather variables were modeled using cubic and piecewise linear splines (Fig. 6).

The non-parametric analysis in Henley and Peirson (1997) show that the typical British household exhibits a reduced responsiveness to temperature below 10 °C (when electric space heating is always on) and above 20 °C (when electric space heating is always off). Moral-Carcedo and Vicéns-Otero (2005) developed a "logistic smooth transition regression" (LSTR) model with a threshold temperature of 15.4 °C. This model has two key advantages: it captures effectively the smooth response of electricity demand to temperature variations in intermediate ranges of temperatures and it provides a method to analyze the validity of temperature thresholds traditionally used to estimate the CDD and HDD variables. Later, Bessec and Fouquau (2008) followed the LSTR method developed by Moral-Carcedo and Vicéns-Otero (2005) and applied it to a panel of 15 member states of the European Union over the last two decades. Fig. 7 shows scatter plots of the filtered electricity consumption against the temperature in three countries representative of warm, intermediate and cold climates: Greece, Germany and Sweden.

As shown in Fig. 7, the relationship between electricity demand and temperature is clearly non-linear in all countries. As expected, heating demand dominates both in cold and intermediate countries such as Sweden and Germany. Warm countries like Greece clearly exhibit both heating and cooling effects. Electricity demand decreases with temperature when it is cold due to the use of heating equipment that is typically fueled by non-electricity sources, while higher temperatures increase the electricity consumption to run cooling devices during warm days. In the case of Greece, there is a zone around 16 °C where the demand is less sensitive to temperature variations.

[3] The Organization for Economic Co-operation and Development.

[4] As fully explained in the Title 24 energy efficiency standards glossary Section 6 (California Energy Commission, 2012), each climate zone has a unique climatic condition that dictates which minimum efficiency requirements are needed for that specific climate zone.

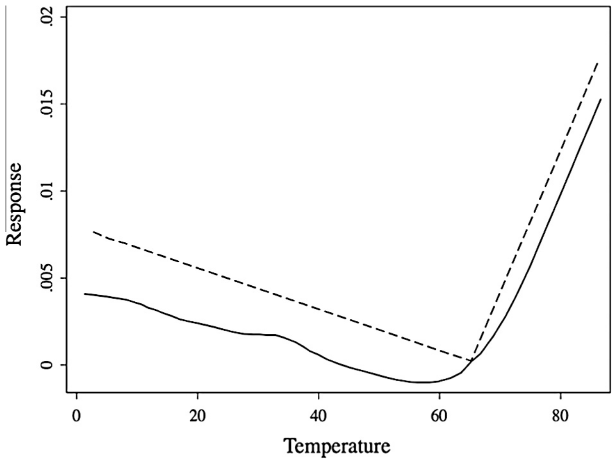

Fig. 6. Replication of Fig. 4 from Engle et al. (1986): Temperature response functions for northeast utilities where the solid curve is the nonparametric estimate and the dashed curve is the parametric estimate.

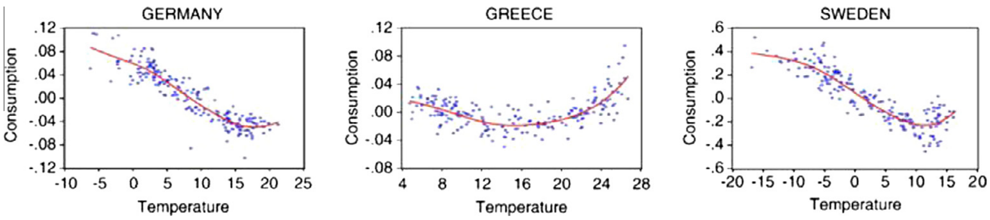

Fig. 7. Replication of Fig. 1 from Bessec and Fouquau (2008): Filtered demand versus temperature. Note: The continuous line depicts a local polynomial kernel regression of order 2 with a Gaussian kernel between the variables.

There are basically two limitations with the application of semi- or non-parametric models. First, these models don't permit detailed policy simulations for changes that potentially affect balance point temperature differentials. Secondly, these models need for their specification larger sample sizes compared to other models as the number of explanatory variables increases.

2.5. Summary of comparisons between models

This review of the recent literature on energy demand responses to temperature shows that researchers attempted to be parsimonious. The simplicity of estimating linear models as well as the ease with which findings can be compared across geographies and time, are still the main drivers for using linear functions. However, several factors violate the linearity assumption, such as constraints associated with the capacity of heating systems and consumer responses to extreme hot and cold temperatures. The application of nonlinear models shows energy demand to be slightly more sensitive to climate than linear models suggested. Non-parametric models, such as logistic smooth transition regression (LSTR), capture the smooth response of energy demand to temperature changes in intermediate ranges of temperatures efficiently. However, these models are more difficult to specify and they require large sample sizes when the number of explanatory variables increases.

2.6. Other driving factors

With a focus on the sensitivity of residential energy demand, researchers have studied the influence of other potential factors on residential energy demand, including occupant characteristics (e.g., income, education, age, household size), energy price for households, physical descriptors of the building (e.g., floor area, home vintage, insulation), as well as human factors (e.g., indoor temperature settings, consuming habits, energy conservation efforts) (Sandberg et al., 2014; Tornberg and Thuvander, 2005; Torres et al., 2015; Yao and Steemers, 2005). While such non-temperature information has often been used to specify energy demand more generally, it has rarely found its way into specifications of TRFs.

The importance of income on the temperature-energy demand relationship was studied by Petrick et al. (2010). As expected, households with higher income have more choices to adjust to temperature changes than low-income households, e.g. by improving insulation or heating systems. A similar observation holds for high and low income countries. If temperatures increase, the decrease in cooling load should be steeper, if a country is comparably richer, which implies that the temperature elasticity of energy demand will increase (in absolute value) with rising income.

To provide more detailed information than the aggregate values regarding occupant characteristics and appliance penetration levels, housing surveys are conducted. These surveys target a sample of the population. Examples are Canada (Office of Energy Efficiency, 2006), USA (Energy Information Administration, 2001), and UK (Department for Communities and Local Government, 2007). Moreover, with the installation of smart meters, interval energy data are more available and can be used to estimate base load, peak load, and weather sensitivity (Granderson et al., 2011; Pedersen et al., 2008; Price, 2010). For example, Kara et al. (2014) have used a dataset collected from over 4200 homes in Texas during the summer of 2012 to develop autoregressive moving average (ARMA) models for individual households. They studied the dependency of household demand response availability to the temperature set point schedule, outdoor air temperature and time of the day. More recently, The Energy Information Administration (EIA) has investigated the potential benefits of incorporating interval electricity data into its residential energy end use models (EIA, 2015).

Another key element that potentially affects the accuracy of temperature response functions is the difference between the outdoor temperature that is felt in buildings and temperature reported at the nearby weather station. The error is more significant in national-scale analyses. For example, Valor et al. (2001) used the mean daily temperatures measured at only four weather stations distributed across Spain, namely, Madrid (central Spain), Bilbao (north), Valencia (east), and Seville (south). Whether these measurements are of direct relevance to energy consumption profiles as specific localities all over Spain, however, is doubtful, given considerable microclimatic variations across the country.

Building energy simulation tools such as DOE 2 and EnergyPlus are increasingly used to assess the climate impacts on building energy demand and energy performance of a given building and thermal comfort for its occupants based on buildings physical characteristics, system efficiency, and operation conditions (Huang, 2006; Loveland et al., 1990; Maile et al., 2007; Scott et al., 2005; 1994). As solar radiation, humidity, and building characteristics such as thermal mass are not considered in degree day analysis, energy simulation tools can be applied to overcome this limitation of temperature response functions and develop hour-by-hour energy simulation to better study the impact of climate change on heating and cooling energy consumption in buildings (Scott et al., 1994). In Table 1, the studies and selected explanatory variables for the reviewed literature are listed.

3. Application of temperature response functions

In the previous section we classified and discussed different types of temperature response functions. In this section, we turn to the application of these functions for energy demand projection. Afterwards, we review their potential to evaluate the impacts of adaptation measures targeted towards lowering climate impacts on household energy demand.

3.1. Energy demand projection

Changes in household energy consumption are driven by the shape of the weather-consumption relationship and the change in projected climate. In the most common approaches, residential energy demand is projected based on regression results for energy carriers in conjunction with climate change scenarios to project future energy use. Amato et al. (2005) assume that other driving factors such as the efficiencies of energy using technologies, household structure and size, as well as household income would continue to change as they have over the analysis period. Their climate change scenario for 2020 estimated a 2.1% increase in per capita residential electricity consumption in Massachusetts (USA) relative to a 2020 non-climate change scenario. In fact, this result is based on the hypothesis that temperature thresholds for calculating HDD and CDD will be stationary, while there is some evidence of acclimatization to warmer conditions which may influence the interpretation of the analysis (Sailor and Pavlova, 2003).

De Cian et al. (2012) used temperature elasticities estimated from a log-linear formulation to project energy consumption levels for a selected group of countries in the year 2085, distinguishing the three energy carriers electricity, gas and oil. Based on the results De Cian et al. (2012) observe that there is a clear heating effect for cold countries and a cooling and heating effect in hot countries. Results are mixed for mild countries. The results showed the capability of the approach to be applied in global climate-economy models.

To project degree-day values into the future, NEMS (U.S. Energy Information Administration, 2013) takes the past decade average and adjusts it by projected shifts in population. Space cooling received an exponential weighting (1.1) to model a non-linear relationship between energy consumption for space cooling and temperature while space heating followed a linear trend. Brown et al. (2015) shows that the projection outcome will be 31% higher when the exponent factor is 1.5. The nonlinear nature of these models makes the estimated parameters highly sensitive to data and model specifications. Besides, uses of nonlinear models for forecasting purposes typically suffer the same limitation as linear models as they assume no change in the temperature response curve.

To the best of our knowledge, there are limited efforts to apply semi- or non-parametric models to project energy demand. Among the exceptions is Moral-Carcedo and Vicéns-Otero (2005), which proposed an approach to resolve the limitation of previous studies that assumed a stationary temperature response curve. They studied the temperature thresholds

Table 1
An overview of the developed models to study the climate change impact on energy demand in buildings.

Authors	Method	Explanatory variables
National studies		
ICF (1995)	Study the impact of higher average daily temperature on electricity demand, and changing conditions on hydroelectric output, thermal power plant performance for six case study utilities	Population, household size, per capita income, electricity price, efficiency, daily average temperature
Rosenthal and Gruenspecht (1995)	Apply a three step approach to estimate the impact of global warming on U.S. energy expenditures for space heating and cooling in residential and commercial buildings.	Average price of fuel, square footage of building, degree days (The recommended balance points, which they used are: commercial cooling 50 °F; commercial heating 50 °F; residential cooling 65 °F; and residential heating 60 °F)
Henley and Peirson (1997)	Estimating the linear, non-linear and non-parametric response functions for a sample of UK households	Monthly mean temperature, electricity price
Lam (1998)	Regression of residential electricity consumption for Hong Kong	Average monthly income per household, household size, electricity price, cooling degree-days (CDD), latent enthalpy-days (LED) and cooling radiation-days (CRD)
Considine (2000)	Econometric modeling of carbon emissions using monthly models of US energy demand due to climate fluctuations	Real price, real disposable income, employment per sector, HDD, CDD (The balance points of 65 °F is used for both heating and cooling)
Vaage (2000)	Discrete/continuous choice approach for Norwegian household based on cross-sectional micro data	Energy price, household gross income, number of rooms in buildings, age of the building, type of building, climate
Mendelsohn (2001)	Econometric analysis of RECS and CBECS microdata for seven regions of the United States	Energy price, market penetration of air conditioning, Precipitation
Pardo et al. (2002)	Econometric forecasting of daily electricity load in Spain	Dummy variables for the days of the week and months, HDD, CDD (The balance points is 18 °C for both heating and cooling)
Zarnikau (2003)	Comparison of linear, log-linear, translog, and non-parametric functions for estimating electricity consumption for heating based on the cross-sectional household-level data	Energy expenditure on electricity, price of electricity, price of natural gas, HDD
Sarak and Satman (2003)	Linear estimation of natural gas consumption for residential heating in Turkey based on HDD with three base temperatures 18.3, 17 and 15 °C	Overall heat transfer coefficient for the building, number of apartments, the efficiency of the heating system and HDD with three base temperatures 18.3, 17 and 15 °C
Scott et al. (2005)	Analysis of energy consumption in U.S. residential and commercial buildings using FEDS and BEAMS models	Building stock, temperature
Moral-Carcedo and Vicéns-Otero (2005)	Logistic Smooth Transition Regression (LSTR) model to analyze Spanish electricity response to temperature variations	Working day effect, Mean daily temperature
Mansur et al. (2005)	Econometric analysis of RECS and CBECS microdata (Discrete-Continuous Choice Model)	Monthly average temperature, fuel price (electricity, fuel oil, natural gas, LPG), number of floors, number of household members, size of building, age of building, head householder age
Fung et al. (2006)	Testing four types of functions (linear, quadratic, cubic and exponential) for energy consumption in Hong Kong	Yearly mean temperature (Six meteorological variables were tested; ambient temperature, relative humility, wind speed, wind direction, solar radiation and precipitation. The results showed that temperature explained about 73% of the total variance)
Hadley et al. (2006)	NEMS energy model, modified for changes in degree-days	HDD and CDD (The reference temperature is 65 °F for both heating and cooling), fuel prices and availability, floor area, equipment retail cost and efficiency, appliance saturation level
Huang (2006)	DOE-2 building energy model combined with the climate changes in four IPCC scenarios for 16 U.S. locations	Building layout, construction, usage, and conditioning systems (lighting, HVAC, etc.), along with weather data (average climate changes are used to modify daily temperature profiles at modeled locations)
Bigano et al. (2006)	Dynamic Panel Analysis of the demand for coal, gas, electricity, oil and oil products by residential, commercial and industrial users in OECD and (a few) non-OECD countries	Real GDP, end-user prices and yearly average temperature
De Cian et al. (2007)	Dynamic Panel Analysis to estimate the short- and long-run temperature demand elasticities for a panel of OECD and non-OECD countries for different types of fuels	Household fuel prices, average seasonal temperature levels, GDP per capita
Bessec and Fouquau (2008)	Logistic transition function (LPSTR model) and an exponential transition function (EPSTR model) for the relationship between electricity demand and temperature for 15 member states of the European Union	Monthly average temperature, population and production in total manufacturing for the period of 1985–2000
Shorr et al. (2009)	Models for heating/cooling expense in 13 states in the northeastern United States	Energy intensity, price of heating energy and electricity, HDD, CDD (The balance point is

(*continued on next page*)

Table 1 (*continued*)

Authors	Method	Explanatory variables
		65 °F for both heating and cooling)
Hekkenberg et al. (2009)	Dynamic temperature dependence patterns in future energy demand	Base daily energy demand, upper threshold temperature for heating demand, lower threshold temperature for cooling demand, HDD, CDD
Isaac and van Vuuren (2009)	Energy as a function of Activity (A), Structure (S) and Energy intensity (I)	Population, floor area, HDD (The base temperature is 18 °C), useful energy heating intensity, efficiency of a representative technology for each energy carrier
Petrick et al. (2010)	Different functional forms (linear, quadratic, polynomials, logarithmic and inverse functions) for fuel demand response to temperature on an unbalanced panel of 157 countries over three decades	Daily mean temperature of the coldest month, the daily mean temperature of the hottest month in one year, income per capita, fuel price, the cross price of substitutable fuels and country specific effect
Hamlet et al. (2010)	Linear relationship between HDD and energy demand in the Pacific Northwest and Washington State	Population, AC Penetration, Annual HDD and CDD (The base temperatures for heating and cooling are 18.33 °C and 23.89 °C, respectively)
Eskeland and Mideksa (2010)	Econometric estimation of residential electricity consumption in 31 countries	Real income per capita, price of electricity, CDDs HDDs (The base temperatures for heating and cooling are 18 °C and 22 °C, respectively), total tax, Value added tax, Value added tax revenue
Cho et al. (2011)	Revise the HDD estimation for 15 major cities of South Korea	Daily average temperature, the indoor set-point temperature and unloaded temperature.
Dowling (2013)	The POLES global energy model (The impacts considered are changes in heating and cooling demand in the residential and services sector, changes in the efficiency of thermal power plants, and changes in hydro, wind (both on- and off-shore) and solar PV electricity output)	GDP/capita, energy prices, technology efficiency improvements, HDD, CDD (The base temperatures for heating and cooling are 15 °C and 18 °C, respectively), penetration of ACs, impact of CDD on AC penetration level
Labriet et al. (2013)	The technico-economic TIMES-WORLD and the general equilibrium GEMINI-E3 model are coupled with PLASIM-ENTS (a climate model)	Price of residential consumption, the aggregated energy price for residential sector, elasticity of substitution, technology shift population weighted HDD and CDD (The base temperature for both heating and cooling is 18 °C)
Jylhä et al. (2015)	Dynamic building energy simulation tool combined with hourly test reference weather data sets for 2030, 2050 and 2100 for a typical detached house in Finland	Hourly weather data (temperature, relative humidity, global solar radiation and wind speed), energy performances of the HVAC systems, price of electricity and district heating, set point temperatures (The set point temperatures during occupied hours were 21.5 °C and 23 °C for heating, depending on the zone and 25 °C for cooling)
Regional studies		
Donovan and Fischer (1976)	Three models for residential heating need in New England (A cross-sectional model that depicts consumption per degree-day as a function of physical and occupant characteristics of a home; A time series regression model that establishes consumption per degree-day as a function of price and consumer awareness of an energy shortage; and a cross-sectional regression model that attempts to explain change in consumption per degree-day from one year to the next as a function of specific conservation actions such as temperature resetting, addition of storm windows)	Number of rooms, year home built, income, presence of insulation, presence of basement, day temperature, night temperature
Loveland et al. (1990)	CALPAS3 Building Energy Model (Six cities and five building types representing a range of climates and building occupancies were modeled)	Building type (Mercantile, General office, large office, single family detached, mobile home), Physical characteristics for buildings (square footage, number of floors, etc), Climate-Dependent Building Characteristics (*U*-values of floor, roof, windows, etc), HDD, CDD
Baxter and Calandri (1992)	Building energy model to forecast electricity demand of the California Energy Commission based on a very detailed dataset	Building design, construction and operation, climate variables, such as annual average temperature, humidity and cloud cover
Sailor and Munoz (1997)	Econometric on state time series to assess the sensitivity of electricity and natural gas consumption to climate at regional scales	Wind speed, relative humidity, enthalpy latent days, HDD, CDD (Degree days are with 18.3 °C base temperature for all states except Florida which requires a degree-day base of 21 °C)
Sailor and Pavlova (2003)	Econometric on state-level time series to analyze the long-term impact of climate change on residential cooling energy demand through increase in the market saturation of air conditioning	Mean monthly wind speed, CDD, HDD (The base temperature for degree-day calculations is 18.3 °C)
Mendelsohn (2003)	Econometric on national cross sectional data on RECS and CBECS data (The national analysis suggests that energy expenditures for both residential and commercial property have a U-shaped relationship with respect to temperature)	Mean temperature of January and July, fuel prices (electricity and natural gas), year built, income, number of floors, family size, age of household, appliances

Crowley and Joutz (2003)	Time series econometric on state data to assess the climate change-driven effects on electrical energy demand and production	Dummy variables for the day of the week and summer months, Holidays, CDD (The base temperature for cooling-degree-day calculations is 72 °F)
Amato et al. (2005)	Time series econometric on state data to assess temperature sensitive energy demand in the Commonwealth of Massachusetts, USA	Fuel prices (electricity, natural gas, heating oil), hours of daylight, HDD, CDD (balance point temperatures for electricity consumption are 60 °F and 55 °F in the residential and commercial sectors, respectively)
Ruth and Lin (2006)	Time series econometric on state data to explore potential impacts of climate change on natural gas, electricity and heating oil use by the residential and commercial sectors in the state of Maryland, USA	Daylight, fuel price, HDD, CDD (The balance point temperature for electricity in the residential and commercial sector are 60 °F and 53 °F, respectively; 71 °F for natural gas used for heating in both the residential sector and commercial sector, and 64 °F for heating oil)
Asadoorian et al. (2006)	Combination of temporal and regional data sets for electricity consumption in urban and rural areas in China	Stock of ACs, electricity price, income per capita, average temperature, price of ACs and fans, residential space area
Giannakopoulos and Psiloglou (2006)	Energy load analysis for the Greater Athens area in Greece.	Mean daily air temperature, HDD, CDD (The base temperature for heating and cooling-degree-day calculations is 22 °C)
Timmer and Lamb (2007)	Linear model relating Temperature and Residential Natural Gas Consumption in the Central and Eastern United States	Days below percentile (DBP) index, population, HDD, (HDD values are calculated using a wide range of reference temperatures (from 0 °C/+32 °F to +40 °C/+104 °F at 2 °C/ 3.6 °F intervals)
Deschênes and Greenstone (2007)	Panel data-based approach to study the (nonlinear) relationship between daily temperatures with annual mortality rates and with annual residential energy consumption	State fixed effects, census division by year fixed effects, population and GDP by state, HDD, CDD (both are calculated with a base of 65 °F)
Franco and Sanstad (2008)	Regression of electricity demand to study the relationships between temperature and both electricity consumption and peak demand at a sample of locations around California	Average daily temperature and maximum hourly temperature
Dubin (2008)	Engineering thermal load model to impute balance point temperatures for residential households in the Puget Sound Energy for their Washington service territory	Number of household members, presence of wall insulation, inches of attic insulation, number of room and floor, floor area, number of windows, annual income, HDD
Miller et al. (2008)	Estimation of extreme cooling demand based on extreme heat days	CDD (The base temperature is 65 °F) , (extreme-heat is defined as the temperature threshold for the 90th-percentile exceedance probability (T90) of the local warmest summer days under the current climate
Auffhammer and Aroonruengsawat (2011)	Panel data-based approach to study flexible temperature response functions for residential electricity consumption in California	Population, household income, mean daily temperature
Zhou et al. (2014)	Detailed buildings energy model with U.S. state-level representation, nested in an integrated assessment framework of the Global Change Assessment Model (GCAM) to project state-level buildings energy demand and its spatial pattern	State level GDP, floor space area, building shell efficiency and surface ratio, satiation impedance per income, population-weighted HDD/CDDs (Degree days are calculated with a set point of 18 °C), "share-weight" of technology
Brown et al. (2015)	Nonlinear model for estimating electricity consumption in residential and commercial buildings	CDD with variable threshold temperature (The best-fitting CDDs have a balance point temperature of 67 °F, for both residential and commercial buildings)

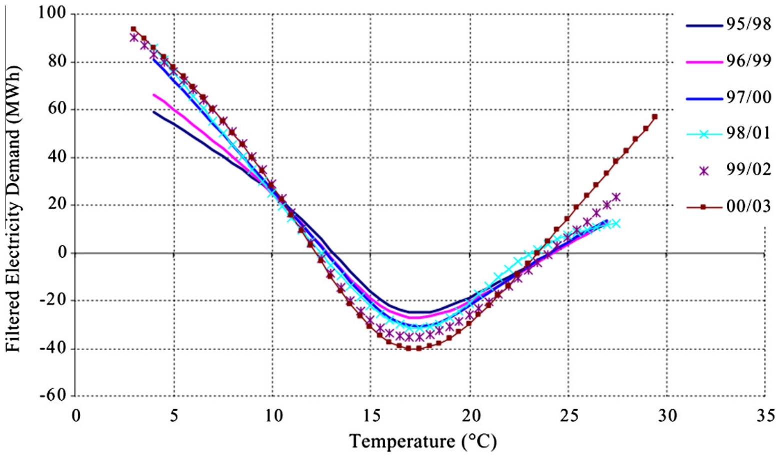

Fig. 8. Replication of Fig. 8 in Moral-Carcedo and Vicéns-Otero (2005): Estimated demand response to temperatures by sample period. LSTR Model.

for a 3-year rolling sample and identified the changes in the slope of demand response and in the value of temperature thresholds (similar to Fig. 8). This noticeable increase in the slope of the demand response in recent years implies that the use and diffusion of heating and cooling equipment installed in Spain are more intensive than before. Besides, it may correspond to changes in the set-point temperature by consumers in response to higher temperatures observed over the last two decades.

3.2. Analysis of adaptation potentials

So far, there are only a few attempts to investigate the potential impact of adaptation measures on future residential energy demand. Miller et al. (2008) recommend raising the balance point temperature for air conditioning by 10 °F as a potential adaptation strategy. They estimated that raising the CDD threshold to 75 °F in San Francisco would restrict the CDD growths to 15 degree days per year by the end of the century, which could remove any increases in projected demand in both high and low emission scenarios.

Subsequently, Auffhammer and Aroonruengsawat (2011) studied the significance of defining different response functions on projecting the electricity demand by testing two scenarios using the response function of coastal San Diego (Zone 7) in an 'almost best case' scenario, and the response function of Central Valley (Zone 12), in the 'almost worst case' scenario. Total increases in simulated electricity consumption in the 'almost worst case' scenario is estimated to be twice of the 'almost best case' scenario. This enormous influence of the assumed temperature response function on overall simulated residential electricity consumption highlights the significance of improving energy efficiency of buildings and appliances.

As described in the previous section, most of the studies that focused on projecting the residential energy demand have assumed that temperature response functions are fixed for each climate zone, which is a critical and rather unrealistic assumption. As a short-term response (intensive margin), people may tend to adjust the thermostat set-point to cope with weather shocks, while in the long-term (extensive margin), they may focus on improving the energy performance of their buildings, installing new heating-cooling equipment, or changing their behavior towards using space conditioning equipment (Auffhammer and Mansur, 2014). To overcome this shortcoming, one approach is to adjust the response function for a cluster of buildings with similar characteristics.

Another common weakness of the previous studies that narrows the credibility of the findings on local adaptation measures in urban area is the unavailability of temperature data very close to the point of energy end-use, as the weather station temperature is used to analyze the potential of different adaptation strategies. The accuracy in identifying adaptation potentials will be significantly enhanced, if data from locally installed sensors are used.

4. Conclusion

This paper provides a comprehensive review of the different applications of baseline temperature, shape and type of the temperature response functions of residential energy demand. Indeed they are used differently considering different climate conditions and perception of temperature response for heating and cooling. The applied models fall into one of four groups – linear, linear asymmetric, nonlinear and semi-or non-parametric. Among them, linear models are the most popular, given the simplicity of estimating these models and the ease with which findings can be compared across geographies and time. However, several factors have been identified that undermine the linearity assumption such as the capacity of heating systems and consumer responses to extreme hot and cold temperatures. Applications of nonlinear models show that energy demand is slightly more sensitive to climate than the estimation from linear models suggests (Brown et al., 2015; EIA, 2001).

Despite the technical difficulties, Moral-Carcedo and Vicéns-Otero (2005) identified two major benefits from applying a non-parametric model – the logistic smooth transition regression (LSTR) model. It captures the smooth response of energy demand to temperature changes in intermediate ranges of temperatures efficiently and at the same time, it can assess the precision of temperature thresholds that are usually used to estimate the CDD and HDD variables.

Our review indicates that the explanatory variables found in the bulk of the literature on energy demand responses to temperature rarely capture the effect of building characteristics including age, type of building and quality of building envelope. Furthermore, we find that the application of models in projecting energy demand rests on the assumption that temperature response functions are stationary, while the evidence of acclimatization to warmer conditions may suggest otherwise. As a resolution, Moral-Carcedo and Vicéns-Otero proposed to study the temperature thresholds for a 3-year rolling sample to pinpoint the source of changes observed in the slope of demand responses and temperature thresholds.

In order to improve understanding of the adaptation potential to climate change, two major demand responses to temperature alteration were identified – changes in the behavior of residents are assumed to occur in response to short-run weather shocks (e.g., by changing the thermostat set-point), and long-term decisions are made about timing of purchases or replacements of durable products. The reviewed studies only tested the impact of changes in the temperature threshold and the shape of temperature response functions. In order to include long-term effects on adaptation potentials, it is necessary to capture building stock turnover rates and associated changes in technology (e.g. passive heating and cooling), as well as behavioral responses of building occupants (acclimatization).

The other limiting factor found in this review that potentially affects the accuracy of temperature response functions is the difference between the outdoor temperature that is felt in buildings and temperatures reported at the nearby weather stations. This is especially an issue in studies conducted at a national level, but also applies at the region- or city-scale. Lack of knowledge about the differences between reported temperatures and the temperatures that are felt in buildings would directly affect the accuracy of projecting residential energy demand as well as the adaptation potential for buildings.

Given these limitations of previous studies, future research will need to address the quality and other properties of temperature data through in situ sensing of temperature and energy demand. Using these data in conjunction with information on the type and age of the building stock should help generate improved assessments of the adaptation potential at various aggregation levels, and enable energy system managers and policy makers to capture the impacts of building replacement strategies, retrofitting policies and building regulations on energy demand in the context of non-stationary climate conditions. Combining that information with data on building occupants, and projecting their changes over time, for example through the use of demographic models, may further enhance energy demand projections.

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